

DONGSU LEE

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EDUCATION & AFFILIATION

University of Texas at Austin

Ph.D. student in Department of Electrical and Computer Engineering
Adviser: Prof. Amy Zhang

Austin, TX
Aug. 2025 - Present

Carnegie Mellon University

Visiting student in School of Computer Science
Adviser: Prof. Ding Zhao

Pittsburgh, PA
Aug. 2024 - Feb. 2025

Soongsil University

Student Researcher
Collaborator: Prof. Minhae Kwon

Seoul, Korea
Mar. 2022 - Feb. 2025

Soongsil University

B.S., Bio-medical System; Big-data Computing
Leave of absence for military service: Aug 2017 - May 2019 (2 years)

Seoul, Korea
Mar. 2016 - Feb. 2022

PUBLICATIONS & PREPRINTS (GOOGLE SCHOLAR)

Conferences

1. **Dongsu Lee**, Daehee Lee, Amy Zhang
Multi-agent Coordination via Flow Matching
International Conference on Learning Representation (ICLR), 2026 
2. **Dongsu Lee**, Daehee Lee, Yaru Niu, Honguk Woo, Amy Zhang, Ding Zhao
Unifying Agent Interaction and World Information for Multi-agent Coordination
NeurIPS ARLET Workshop, 2025 (Oral) 
3. Daehee Lee, **Dongsu Lee**, TaeYoon Kwack, Wonje Choi, Honguk Woo
Policy Compatible Skill Incremental Learning via Lazy Learning Interface
Conference on Neural Information Processing System (NeurIPS), 2025 (Spotlight) 
4. **Dongsu Lee**, Minhae Kwon
Temporal Distance-aware Transition Augmentation for Offline Model-based Reinforcement Learning
International Conference on Machine Learning (ICML), 2025 
5. **Dongsu Lee**, Minhae Kwon
Episodic Future Thinking Mechanism for Multi-agent Reinforcement Learning
Conference on Neural Information Processing System (NeurIPS), 2024 
6. **Dongsu Lee**, Chanin Eom, Minhae Kwon
AD4RL: Autonomous Driving Benchmarks for Offline Reinforcement Learning with Value-based Dataset
IEEE International Conference on Robotics and Automation (ICRA), 2024 

Journals

1. **Dongsu Lee**, Minhae Kwon
Scenario-free Autonomous Driving with Multi-task Offline-to-online Reinforcement Learning
IEEE Transactions on Intelligent Transportation Systems, 2025. 
2. **Dongsu Lee**, Minhae Kwon
Offline Reinforcement Learning on Foresighted Decision-making for Autonomous Driving System
IEEE Internet of Things Journal, vol.12, no.19, 2025. 
3. **Dongsu Lee**, Minhae Kwon
Instant Inverse Modeling of Stochastic Driving Behavior with Deep Reinforcement Learning
IEEE Transaction on Consumer Electronics, September 2024 
4. Chanin Eom, **Dongsu Lee**, Minhae Kwon
Selective Imitation for Efficient Online Reinforcement Learning with Pre-collected Data
ICT Express, vol. 11, no. 2, 2025 

5.

Jeyeon Eo, **Dongsu Lee**, Minhae Kwon

The Impact of Dataset on Offline Reinforcement Learning Performance in UAV-based Emergency Network Recovery Tasks

IEEE Communication Letters, vol. 28, no. 5, May 2024



6.

Dongsu Lee, Minhae Kwon

Stability Analysis in Mixed-Autonomous Traffic with Deep Reinforcement Learning

IEEE Transactions on Vehicular Technology, vol 72, no. 3, March 2023

NeurIPS 2021 Deep Reinforcement Learning Workshop



7.

Dongsu Lee, Minhae Kwon

ADAS-RL: Safety Learning Approach for Stable Autonomous Driving

ICT Express, vol.8, no.3, September 2022



PATENTS

Issued US Patents

1.

Chanin Eom, **Dongsu Lee**, Minhae Kwon, “Traffic Control Method and Apparatus with Autonomous Vehicle based on Adaptive Reward,” US Patent, Jan 4, 2024. (Application no. US 18/404077), issued on November 18, 2025. (Patent no. US 12,475,785)

2.

Dongsu Lee, Minhae Kwon, “Method and Apparatus for Inferring Stochastic Driving Characteristic of a Vehicle,” US Patent, April 26, 2023. (Application no. US 18/139602), issued on December XX, 2025. (Patent no. US XX)

3.

Dongsu Lee, Minhae Kwon, “Apparatus and Method for Inferring Driving Characteristics of Vehicle,” US Patent, Oct 4, 2022. (Application no. US 17/959515), issued on Aug 19, 2025. (Patent no. US 12,391,261)

4.

Dongsu Lee, Minhae Kwon, “Method for Combating Stop-and-Go Wave Problem Using Deep Reinforcement Learning based Autonomous Vehicles, Recording medium and device for performing this the method,” US Patent, Nov 27, 2021. (Application no. US 17/535567), issued on September 17, 2024. (Patent no. US 12,091,025)

HONORS AND AWARDS

Scholarships

• **Hyundai Overseas Scholarship**

Hyundai Motor Chung Mong-Koo Foundation

2025

• **UT Austin Engineering Fellowship**

Cockrell School of Engineering and The University of Texas at Austin Graduate School

Aug 2025 - Present

• **Hyundai Future Industrial Talent Scholarship**

Hyundai Motor Chung Mong-Koo Foundation

Aug 2023 - Feb 2025

• **Soongsil Graduate Admission Excellence Scholarship (Tuition)**

Full tuition, insurance, and living expenses support for graduate studies

2022

• **Yesan Fellowship Local Talent Scholarship**

2021

• **Soongsil Academic Excellence Scholarship (Tuition)**

2016, 2019, 2020

Awards

• **Global Excellence Scholarship** (three times)

Hyundai Motor Chung Mong-Koo Foundation

2024

• **Best Journal Paper Award**

Korean Institute of Communications and Information Science

2023, 2024

• **ICT Challenge IITP President’s Award**

Ministry of Science and ICT and Institute for Information & communication Technology Planning & evaluation

2023

• **Haedong Best Paper Award**

Korean Institute of Communications and Information Science

2023, 2024

• **Best Paper Award**

Korean Institute of Communications and Information Science

2022, 2024

- **ICT Idea Competition Award** 2021
Korean Institute of Communications and Information Science

Grants

- **NeurIPS Travel Grants, Vancouver, Canada** 2024
Neural Information Processing Foundation
Full conference registration and accommodation support
- **AI Intensive Program at Carnegie Mellon University, Pittsburgh, PA** 2024 - 2025
IITP & Sogang University
- **ICRA Travel Grants, Yokohama, Japan** 2024
IEEE Robotics and Automation Society
- **18th Qualcomm IT Tour, San Diego, CA** 2022
Qualcomm Incorporation
Full support including flight, accommodation, and living expenses
- **Upper Bound: AI Conference, Edmonton, Canada** 2022, 2024
Alberta Machine Intelligence Institution (AMII)
- **Conference Financial Assistance, Virtual** 2021
Neural Information Processing Systems (NeurIPS)

EXPERIENCES

Professional Experiences

- **ICML Theory of Mind Workshop** (Program Committee) Jul 2023
- **EIRIC Rising Researcher Seminar** (Presenter) Electronic & Information Research Information Center Jun 2023
- **R&D Promising Technology Showcase** (Presenter) Korea evaluation institute of industrial technology May 2023

Teaching Assistance

- **Multi-media Converging Technology**, Soongsil University 2023 Spring
- **Programming and Practice**, Soongsil University 2021 Autumn

Reviews

- **Journal:** Nature Communication, IEEE TCYB, IEEE TITS, IEEE TCE, IEEE IoTJ, IEEE TVT
- **Conference:** ICLR, ICML, NeurIPS, CoRL, ICRA, IROS, IEEE IV
- **Workshop:** NeurIPS ARLET (2025), ICML ToM (2023), NeurIPS ML4AD (2022)